

SeokHyeon (Lucas) Kong

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RESEARCH INTERESTS

Quantum Computer Architecture, Quantum Error Correction and Detection, Quantum Compiler, Quantum Controller, Quantum Programming Language, Quantum Algorithm Optimization

EDUCATION

Sungkyunkwan University (SKKU)

Bachelor of Science in Electronic and Electrical Engineering
Bachelor of Science in Advanced Semiconductor Engineering

- Cumulative GPA: 3.95 / 4.0 | Major GPA: 4.0 / 4.0

Seoul, South Korea

Expected February 2026
System Architecture Track

The Pennsylvania State University

Exchange Student in Computer Engineering

- Grade: 4.0 / 4.0

University Park, PA

Fall 2024

PUBLICATIONS & PRESENTATION

S. H. Kong, D. Kim, K. Maeng*, and E. Seo*, “Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy,” *IEEE Computer Architecture Letters*, under review, 2025

S. H. Kong, C. Park, J. Moon, and D. Dahl*, “Solving Constraint Satisfaction Problems with Variational Quantum Algorithm on NISQ Devices,” *Poster presented at the Quantum Information Research Support Center (Qcenter)*, Suwon, South Korea, Sep. 2025

S. H. Kong and H. Oh*, “Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system,” *Journal of the Korean Institute of Information and Communication Engineering*, vol. 27, no. 10, pp. 1208–1213, Oct. 2023

RESEARCH EXPERIENCE

IonQ

Research Mentee, Advised by Dr. Denny Dahl

Remote

08/2025–09/2025

- Addressed the Instant Insanity problem on NISQ devices by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods.
- Applied a Variational Quantum Algorithm to the Four Corners Map Coloring problem.

The Pennsylvania State University

Undergraduate Researcher, Advised by Prof. Kiwan Maeng

University Park, PA

08/2024–09/2025

- Conducted in-depth GPU profiling with Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differentially private training.
- Proposed an optimization that resolved the dominant Geometric operation overhead, reducing it from 50% to 19% while preserving Gaussian noise quality.

Arch Lab at SKKU**Suwon, South Korea***Undergraduate Research Intern, Advised by Prof. Dongmoon Min**07/2025–08/2025*

- Built foundational expertise in Quantum / Cryogenic Computer Systems through literature reviews, technical presentations and academic defenses.
- Investigated quantum computer systems, including superconducting-qubit design, error mechanisms, control and readout techniques as well as cryogenic computer architectures.

POSTECH QCQN Lab**Pohang, South Korea***Undergraduate Research Intern**01/2025*

- Performed 397 nm fluorescence (Doppler) and 729 nm quadrupole spectroscopy on Ca^+ ions and enhanced system stability through targeted adjustments.
- Implemented micromotion optimization via 2D scanning with DC compensation rods.

REFERENCES

Kiwan Maeng, Ph.D. (Advisor)*Assistant Professor, Computer Science and Engineering, The Pennsylvania State University*

- Email: kvm6242@psu.edu

Denny Dahl, Ph.D. (Advisor)*Customer Success Director, IonQ*

- Email: denny.dahl@ionq.co

HONORS & AWARDS

NSF Graduate Research Fellowship Program (GRFP)

Applied

DOE Computational Science Graduate Fellowship (CSGF)

Applied

Quantum Hackathon – Director’s Award

*2025*President’s List (Honor Society at SKKU, 2 consecutive years)*2024, 2025*Dean’s List (5 consecutive semesters)*SP2023, FA2023, SP2024, FA2024, SP2025*

Korea-U.S. Student Exchange Scholarship in the field of high-tech industry (\$9,000)

*FA2024***TEACHING EXPERIENCE**

Academic Tutor, Sungkyunkwan University (SKKU)**Suwon, South Korea***Intro to Electromagnetism**Fall 2023***Teaching Assistant**, Sungkyunkwan University (SKKU)**Seoul, South Korea***Engineering Mathematics 2**Summer 2023***TECHNICAL SKILLS**

Software: Qiskit, CUDA, C, Python, MATLAB, LaTeX**Hardware:** System Verilog, Verilog, Verdi, Virtuoso, Nsight Systems, Nsight Compute**CERTIFICATION**

Qiskit Global Summer School 2025 – Quantum Excellence (IBM Quantum)